

at Cornell University took strontium titanate and deposited it on silicon—the main component of most semiconductors. Schlom and his team then integrated circuits in such a way that the silicon squeezed it into a ferro-electric state. For computer users, it could mean no more waiting for the operating system to go online or to access memory from the hard drive.

“Several hybrid transistors have been proposed specifically with ferroelectrics in mind,” said Schlom. “By creating a ferro-electric directly on silicon, we are bringing this possibility closer to realisation.”

New method could lead to narrower chip patterns

MIT researchers have found a new method for etching extremely narrow lines on a microchip. They achieved this breakthrough by using a material that can be switched from transparent to opaque, and vice versa, just by exposing it to certain wavelengths of light. The technique, called ‘absorbance modulation,’ makes it possible to create lines that are only about one-tenth as wide as the wavelength of light used to create them. Part of the trick was to find a suitable photochromic material whose clear and opaque parts would remain stable after the initial exposure to light.

Using this method, the team produced lines just 36 nanometres wide, and claims that it could also place many such lines spaced a similar distance apart. Producing such fine lines is crucial to many new technologies, from microchip manufacturing that is constantly seeking ways to cram more components onto a single chip, to a whole host of emerging fields based on nanoscale patterns. Hitherto, these technologies have faced fundamental limits because they tend to rely on light to produce these patterns, and most techniques cannot produce patterns much smaller than the wavelengths of light itself. The method developed by MIT overcomes that limit.

The team is pursuing possible use of the technique for imaging systems, which could enable new kinds of mi-

croscopes for observing at nanoscale resolution, with possible applications in biology and in materials science. At the same time, they are pursuing ways of using the technique to create even smaller patterns, down to the scale of individual molecules.

New atom processors

Intel has announced two new Atom processors for mobile Internet devices (MIDs)—the Z550 and Z515. The Z550 extends the performance of the MID product line to 2 GHz with Intel Hyperthreading technology,



Intel Atom processors

ogy, setting a new standard for the highest performance processor in the under-3-watt power category. The

Z515 incorporates the new Intel burst performance technology, which enables it to run at 1.2 GHz when performance is needed in existing small and sleek MID form factors.

These new Atom processors, as per Intel, further extend customer choices to enable the best Internet experience in pocketable MIDs.

Anand Chandrasekher, senior vice president and general manager, Ultra Mobility Group, Intel, gave the first live demo of the company's next-generation Atom-based MID platform, codenamed ‘Moorestown.’ He provided a sneak peek into the low-power innovation of the platform by showcasing a greater than 10x idle power reduction compared to today's Atom-based platform in a side-by-side demo. This reduction is made possible through a combination of new power management techniques, a new partition optimised for the MID segments

and Intel's Hi-k 45nm manufacturing process.

World's first 1394 automotive chip

Fujitsu Microelectronics Asia claims to have launched the world's first automotive controller IC that realises high-definition (HD), 1280x720 video transmission over the IDB-1394 in-vehicle multimedia network protocol. The new IC, the MB88395, can simultaneously transmit multiple streams around the vehicle, such as HD video from Blu-Ray DVDs, digital TV, audio and car navigation images. It achieves this by utilising a high-speed 800Mbps physical layer as well as Fujitsu's SmartCODEC that provides high compression and can transmit HD video without perceptible lag.

According to the company, this reduces the system cost of in-vehicle multimedia networks by a maximum of 30 per cent, while reducing the number of wire harnesses by a maximum of 70 per cent to reduce vehicle weight and improve fuel efficiency.

Control your browser with your face

To make your browsing faster, Opera Software has developed a browser with face recognition. Anyone with a webcam can control his Opera 10 browser by moving only his face.

The Face Gestures technology lets you perform frequent browsing operations with natural and easy-to-make face gestures. By using an internal technology dubbed ‘Face Observation Opera Language,’ it recognises pre-determined facial expressions and matches them to commands on the Opera browser. To activate Face Gestures, you need to ensure that your webcam is plugged in and hold down the F8 key.

Opera Face Gestures include more than 45 controls, from opening new tabs to restoring sessions to navigating

